

Sulfur ($S(^3P)$) Reaction with Conjugated Dienes Gives Cyclization to Thiophenes under Single Collision Conditions

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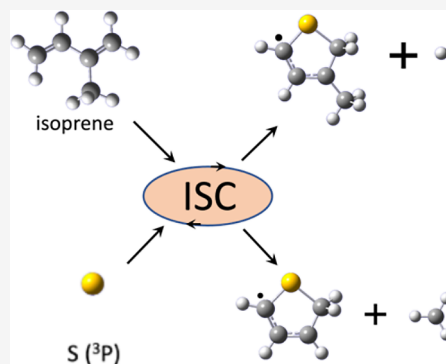


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Supporting Information

ABSTRACT: We combine crossed-beam velocity map imaging with high-level *ab initio*/transition state theory modeling of the reaction of $S(^3P)$ with 1,3-butadiene and isoprene under single collision conditions. For the butadiene reaction, we detect both H and H_2 loss from the initial adduct, and from reaction with isoprene, we see both H loss and methyl loss. Theoretical calculations confirm these arise following intersystem crossing to the singlet surface forming long-lived intermediates. For the butadiene reaction, these lose H_2 to form thiophene as the dominant channel, H to form the detected 2H-thiophenyl radical, or ethene, giving thioketene. For isoprene, additional reaction products are suggested by theory, including the observed H and methyl loss radicals, but also methyl thiophene, thioformaldehyde, and thioketene. The results for $S(^3P)$ + 1,3-butadiene, showing direct cyclization to the aromatic product and yielding few bimolecular product channels, are in striking contrast to those for the analogous $O(^3P)$ reaction.



Sulfur, chiefly as SO_2 , is a common pollutant both as a byproduct of fossil fuel combustion and from volcanic emissions.¹ It has been proposed as part of a geoengineering approach—stratospheric aerosol injection—to increase the Earth's albedo and mitigate global warming.² UV photodissociation of SO_2 has recently been found to give $S(^3P) + O_2$ directly, with significant implications for the early atmosphere of Earth.³ The astrochemical fate of sulfur has long been a puzzle, with high abundances in diffuse interstellar clouds not maintained as they condense into dark clouds.⁴ Recent modeling now points to the difference being organosulfur species frozen onto icy grains.⁵ Organic sulfur species, particularly thiophenes, have also recently been detected on Mars,⁶ stimulating speculation about possible association with biological processes there.⁷ Given these current concerns, we have recently initiated detailed dynamics studies of sulfur atom reactions, supported by state-of-the-art *ab initio* calculations.

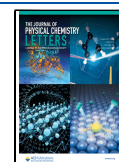
The reaction dynamics of atomic sulfur have been examined both by theory and experiment in recent years, but the focus has exclusively been on the metastable singlet state and in particular its reaction with molecular hydrogen and its isotopologues.^{8–13} In contrast, no crossed-beam studies involving ground state sulfur atoms have been reported while interest in combustion modeling has driven countless crossed-beam studies of $O(^3P)$ reactions with both saturated^{14–17} and unsaturated hydrocarbons.^{18–22} The former give OH by direct abstraction over significant barriers, while the latter are the source of rich chemistry with complex branching that is modulated by the extent of intersystem crossing (ISC) to the lower singlet potential energy surface (PES).

Although to our knowledge no detailed dynamics studies of $S(^3P)$ have been published, both kinetics and product branching have been examined in an extensive series of papers by Gunning and Strausz at the University of Alberta beginning in 1962.^{23–29} One of their more remarkable findings was that addition of $S(^3P)$ across a double bond is highly stereospecific, just as for $S(^1D)$ and singlet species in general.^{26,30} It had been thought that gas phase stereospecific addition would require concerted formation of bonds to both carbon atoms, which is not permitted for the triplet. Gunning and co-workers attributed stereospecificity in this case to asymmetric addition to form a long-lived triplet biradical that yet possesses a high barrier to rotation about the C–C bond.²⁷ However, based on MP2 calculations, McKee suggested instead that the barrier is not high; rather, ISC is very rapid in this system owing to location of the relevant orbitals on the S atom and that the triplet and singlet biradicals are “isoenergetic and isogeometric”.³¹ In one study, directly related to the subject of the present paper, Gunning and co-workers examined $S(^3P)$ reaction with C4 olefins including 1,3-butadiene (13BD) with detection by GC-MS.²⁶ The 13BD reaction, conducted at room temperature and 200–600 Torr total pressure, gave rise mainly to vinyl thiacyclop propane (VTCP), thiophene, and H_2 .

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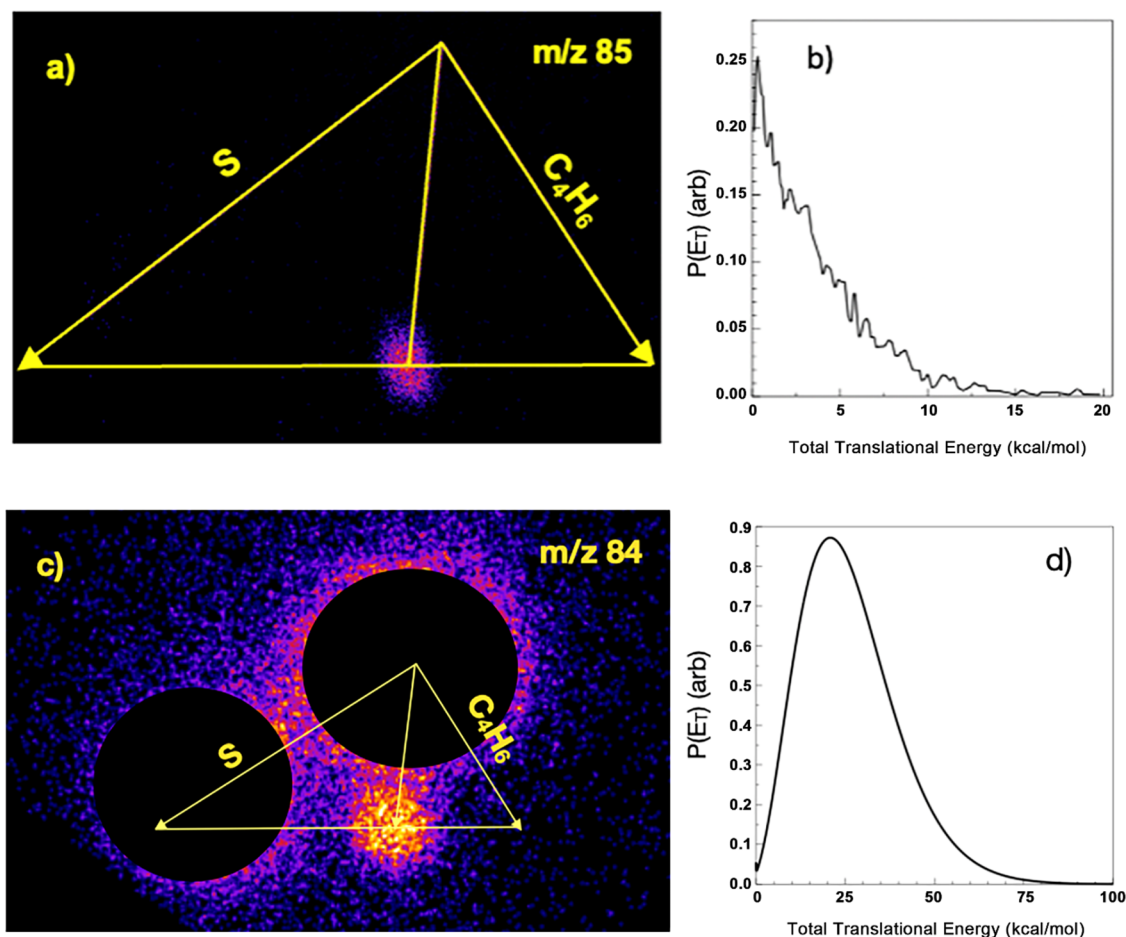


Figure 1. (a) Velocity map image of the 85 u product (C_4H_5S) of the crossed-beam reaction of $S(^3P)$ with 1,3-butadiene at 13.6 kcal/mol collision energy. The most probable Newton diagram is superimposed on the image showing the beam velocities and the center-of-mass velocity of the system. (b) Total translational energy distribution determined for the $S + 13BD \rightarrow C_4H_5S + H$ reaction based on the measured recoil velocity distribution in the image. (c) Velocity map image of the 84 u product (C_4H_4S) from the same reaction with a superimposed Newton diagram. Intense background signals at the origin and the S atom beam have been masked. (d) Total translational energy distribution from analysis of the image in (c).

Much of the thiophene was attributed to subsequent photodissociation of the VTCP product, but a yield of 9% was ascribed to a primary reaction giving thiophene + H_2 as discussed further below.

Here we report the first crossed molecular beam studies of $S(^3P)$ reactions, focusing on conjugated dienes 13BD along with preliminary results for isoprene (2-methyl-1,3-butadiene). We present velocity map imaging^{32,33} studies employing both soft single photon ionization of the radical products at 157 nm and a loosely focused laser to achieve 1 + 1 ionization as described further in the [Supporting Information](#). The results are illuminated with *ab initio* calculations of the initially accessed triplet PES and the singlet PES that leads to products and by RRKM calculations of the branching. The results are in striking contrast to the behavior of the analogous $O(^3P) + 13BD$ system recently studied by Cavallotti et al.,³⁴ and the differences allow for an investigation and discussion of their origin.

In [Figure 1](#) we show results for crossed-beam imaging of the $S(^3P)$ reaction with 13BD at a collision energy of 13.6 ± 2.5 kcal/mol. [Figure 1a](#) gives the velocity map image of the H loss radical product we detect with high sensitivity by single photon ionization, which is at 85 u, C_4H_5S . This corresponds to sulfur addition/H elimination and hence the small size of the image

that appears just at the center-of-mass velocity of the system. The image is shown with the most probable Newton diagram superimposed, confirming that it is the result of the reaction between the S atom and 13BD. [Figure 1b](#) shows the total translational energy distribution extracted from the image, which peaks near zero, extends to ~ 10 kcal/mol, and gives an average translational energy release of 3.7 kcal/mol. [Figure 1c](#) shows the corresponding image for product mass 84 u, thiophene. We searched for this following the theoretical predictions presented below that indicated it is the dominant channel. In this case the ionization energy is slightly above the energy of our probe laser. We thus focused the probe to drive 1 + 1 ionization through Rydberg states. Given the low signal and high background in this case, we used a forward convolution analysis rather than direct inversion to derive the translational energy distribution shown in [Figure 1d](#), as described in the [Supporting Information](#). The translational energy distribution for this closed shell product channel is very different from that for the radical product. It peaks at 24 and extends to 75 kcal/mol. This is clear evidence of an exit barrier as discussed further below.

We also examined the reaction between $S(^3P)$ and isoprene at 15 kcal/mol, and the results are shown in [Figure 2](#). In this case, in addition to the analogous S addition/H elimination

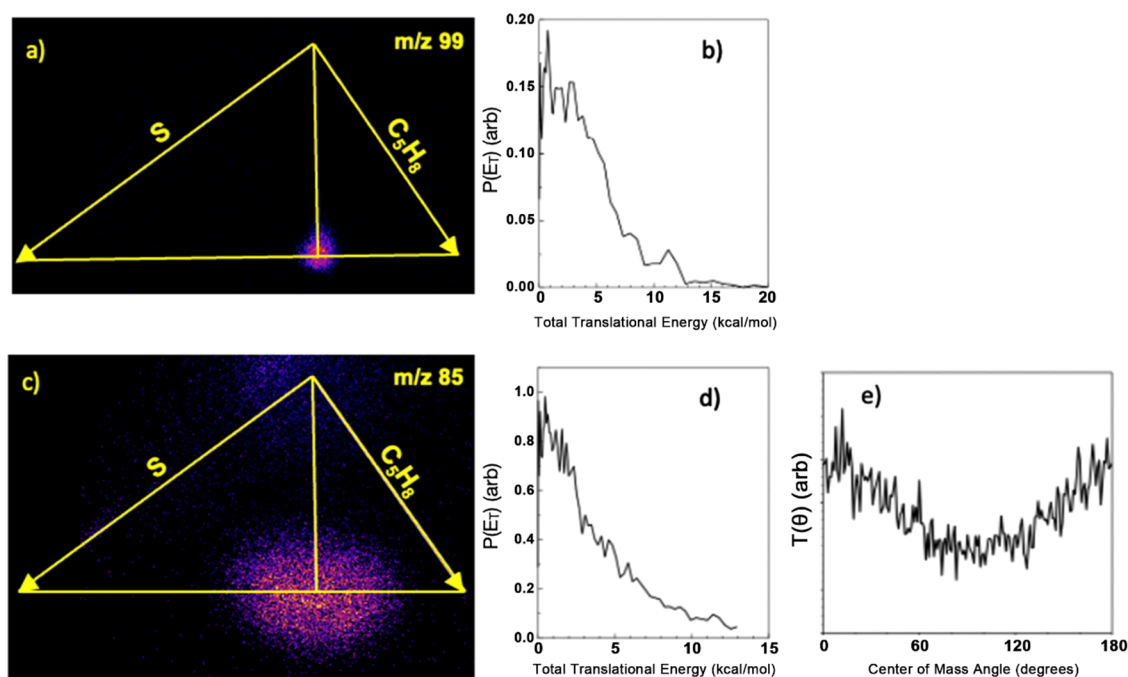


Figure 2. (a) Velocity map image of the 99 u product, C_5H_7S , from the reaction of $S(^3P)$ with isoprene at 15 kcal/mol collision energy with the most probable Newton diagram superimposed. Total translational energy (b) distribution for the $S + \text{isoprene} \rightarrow C_5H_7S + H$ reaction extracted from the image in (a). (c) Velocity map image of the 85 u product, C_4H_5S , from the same reaction. Total translational energy (d) and angular (e) distributions for the $S + \text{isoprene} \rightarrow C_4H_5S + CH_3$ reaction extracted from the image in (c).

channel, which is summarized in Figure 2a,b, we also see S addition/ CH_3 elimination, summarized in Figure 2c–e. The H elimination strongly resembles the 13BD result with an average translational energy release of 4.1 kcal/mol, while the recoil velocity is much larger for the methyl elimination. The low recoil velocity makes it difficult to measure reliable angular distributions for H loss, but for methyl loss this is not the case, and we can extract both translational energy and angular distributions as shown in Figure 2d,e, respectively. The latter is clearly forward–backward symmetric indicating the presence of an intermediate that lives much longer than a rotational period. The participation of a long-lived intermediate decaying to radical products is also suggested by the translational energy distributions that peak near zero.

To identify the reaction products and mechanism, we first focused on the 13BD case. We used KinBot^{35–37} to explore the upper triplet and lower singlet PESs for the reaction that starts at $S + 13BD$ at the CCSD(T)-F12a/cc-pVTZ-F12// $\omega B97X-D/6-311++G(d,p)$ level of theory as described further in Supporting Information. Note that the F12 basis set systematically contains tight d functions that significantly accelerate convergence.^{38,39} Key stationary points on these surfaces are shown in Figure 3. The most relevant minimum found on the initially accessed triplet PES is the trans terminal adduct (3TA), which is 16.1 kcal/mol below the entrance on the triplet PES. The analogous central adduct (3CA) is at -1.6 kcal/mol. The entrance barrier to 3TA is submerged, while that of 3CA is not. We confirmed the existence of a prereactive complex that is stable at the DFT level and is just slightly above the submerged barrier when applying the coupled cluster correction. The two primary adducts, 3TA and 3CA , can interconvert via a loose transition state at 3.3 kcal/mol, but given the low barrier to 3TA and the high barriers controlling further reactions from either of the adducts, no other channel

found on the triplet PES plays a significant role (see also SI, Figure S5). Furthermore, the translational energy release shown in Figure 1b, with an average of 3.7 kcal/mol, is not consistent with formation of a product with <1 kcal/mol energy available. It thus appears that there are no open channels on this PES in the investigated energy range beyond the adducts. This means that (a) no chemistry beyond the adducts can happen on the triplet PES in our experiments and (b) intersystem crossing to the singlet surface is driving the further chemical evolution.

To investigate ISC for this system, we searched for minimum energy crossing points (MECPs) at the $\omega B97X-D/6-311++G(d,p)$ level as described in Supporting Information. We considered the possibility of crossing at six distinct regions, and we successfully located each MECP. The geometries and electronic energies relative to that of the entrance are shown in Supporting Information. We found that the most likely region for ISC is near the 3TA geometry with a SOC strength of 145 cm^{-1} (calculated as the root-mean-square of the SOC elements). We investigated the possibility of prompt rearrangement upon ISC using 100 fs trajectories^{40,41} with 130 kcal/mol kinetic energy for the MECPs near the cis- and trans- 3TA wells as detailed in the Supporting Information, and found evidence of some dynamic effects. While the most important prompt configuration is 1VTCP upon ISC, in $\sim 6\%$ of the cases, other products can form, mostly 1,3-butadiene-1-thiol.

KinBot explored the chemistry on the singlet surface starting from 12SDHT . This species is -72.6 kcal/mol relative to the triplet entrance. We did not find a stable open-shell singlet version of the 3TA structure. 1VTCP , the species associated with ISC, is -58.1 kcal/mol on the same scale and is separated from 12SDHT by a barrier that is below the triplet entrance channel. 1VTCP can further isomerize to 123DHT as well as

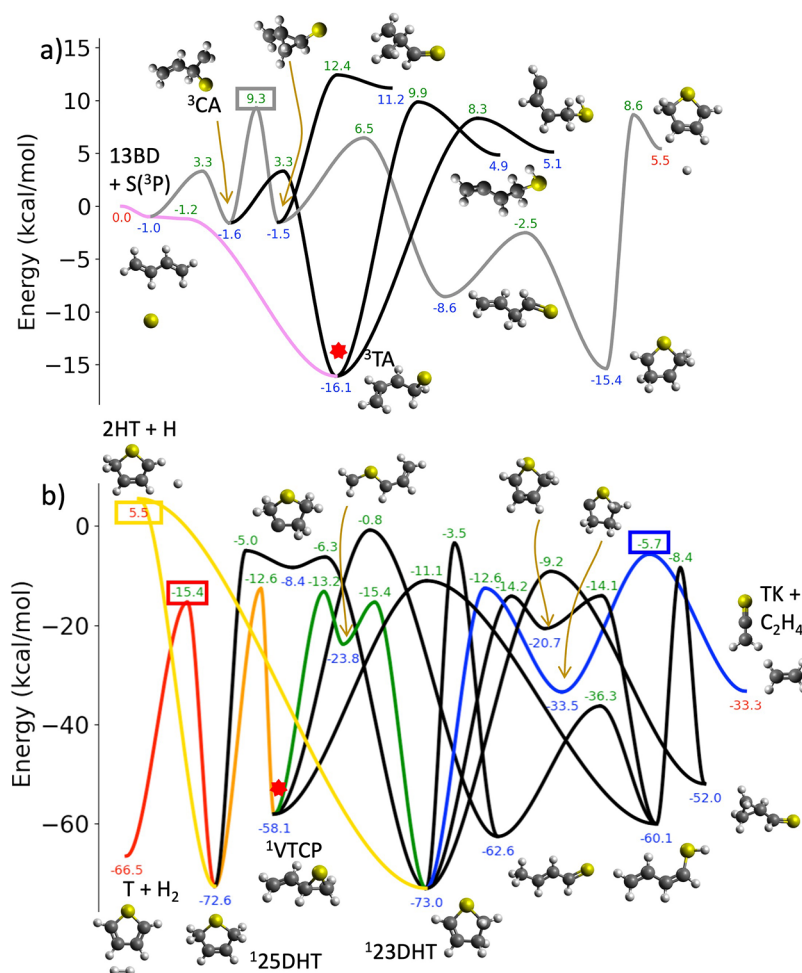


Figure 3. Stationary points on the S + 13BD PESs. Energies are relative to the S(3P) + 13BD entrance at the CCSD(T)-F12a/cc-pVTZ-F12// ω B97X-D/6-311++G(d,p) level and include ZPE. Well energies are shown in blue, barriers are shown in green, and bimolecular product energies are red. Panel (a) shows the part of the triplet PES that is accessible experimentally starting from the addition step. The pink line shows the favored path, while the gray one shows the only open channel in this energy range which leads to bimolecular products with a bottleneck of 9.3 kcal/mol. The ISC location is marked with a star. Panel (b) shows the lowest energy region of the singlet PES upon crossing near the geometry of the terminal adduct, 3TA leading mostly to 1VTCP , marked with a star. To ease interpretation, we removed all wells found by KinBot that do not connect to further species, and the colors show the minimum energy paths leading to the various bimolecular products from 1VTCP , with the bottleneck transition states marked.

thials and thiols. There are very few accessible bimolecular channels. $^{12}SDHT$ can eliminate an H_2 molecule via a tight transition state from the two carbons neighboring the S atom leading to thiophene (T), or eject an H atom to give the resonantly stabilized 2-H-thiophen-5-yl ($2HT$) radical. Interestingly, $^{12}3DHT$ cannot eliminate H_2 in the investigated energy range, but as expected, it can also lead to 2HT . However, this species can, through a carbene intermediate, form thioketene (TK) and C_2H_4 .

To determine the branching fractions following ISC on the singlet PES, we carried out RRKM calculations using the calculated energies and geometries (CCSD(T)-F12a/cc-pVTZ// ω B97X-D/6-311++G(d,p)) amended with 1-D hindered rotor scans to account for anharmonicities (for further details, see [Supporting Information](#)) as implemented in the MESS code⁴² (see [Supporting Information](#) for input). The H atom loss channels from $^{12}3DHT$ and $^{12}5DHT$ were modeled using phase-space theory by setting the high-pressure limit capture rate coefficient of $H + 2HT$ to 10^{-10} cm³/molecule/s with a 1:1 branching ratio into the two DHT species. We used

the marked bottleneck transition states to calculate microcanonical rate coefficients toward bimolecular products starting from 1VTCP . Note that despite being lower, it is the second barrier between 1VTCP and $T + H_2$ that is the rate determining saddle in the energy region of interest (see also SI, [Figure S6](#)).

Our results using the bottleneck barriers are shown in [Figure 4](#) (the branching fractions including an extended model are shown in [Figure S7](#) in the SI). Our calculations indicate that the thiophene:thioketene:2-H-thiophen-5-yl ratio at the energy of the reactants in the crossed-molecular beam experiments (dashed vertical line at 13.6 kcal/mol) is approximately 460:110:1. Although the radical product here is a minor channel, it can be seen clearly in the experiment both because of the sensitivity and selectivity of the direct single-photon ionization and because the H loss product is confined to a very localized region at the center-of-mass of the scattering image. At energies below the 5.5 kcal/mol H loss threshold, of course, the H atom channel is closed. This is consistent with the observations of Strausz and Gunning mentioned in the

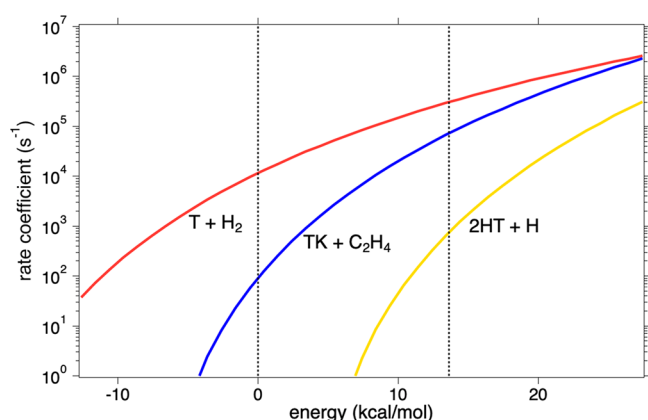


Figure 4. RRKM unimolecular rate coefficients on the singlet PES starting from $^1\text{VTCP}$ and leading to bimolecular products. The zero energy and the notation are the same as in Figure 3. The dashed vertical lines show the energy of the $\text{S}(^3\text{P}) + 13\text{BD}$ asymptote and the additional translational energy in the crossed-molecular beam experiments. See also Figure S6.

introduction.²⁶ At room temperature and high pressure, stabilization to $^1\text{VTCP}$ would be likely and H loss would not be competitive with H_2 loss to form thiophene directly as they observed. At the energy of the $\text{S}(^3\text{P}) + 13\text{BD}$ asymptote (the dashed vertical line at 0 kcal/mol in Figure 4), the T:TK ratio is approximately 130:1.

The experimental translational energy distribution shown in Figure 2d is entirely consistent with the theoretical results in Figure 3b. The available energy is ~ 80 kcal/mol and the saddle point is ~ 50 kcal/mol above the products. Much of this energy is likely to be converted to relative translation owing to Pauli repulsion between the newly formed closed-shell products, hence the peak far from zero. Although thiophene is the major predicted pathway, the weak signals observed for thiophene relative to 2HT arise from the different probe mechanisms: Direct single photon ionization for the radical, and 1 + 1 for thiophene.

The experimental results suggest analogous behavior for isoprene, and this is supported by a preliminary investigation of the PES for this system. For isoprene, methyl elimination giving the same 2HT radical is 4.3 kcal/mol exoergic. In contrast, H elimination to give 2H-3methyl-thiophen-5-yl or 2H-4methyl-thiophen-5-yl radicals is 3.8 and 4.3 kcal/mol endoergic, respectively. H elimination on the triplet PES also can give acyclic radical products analogous to the 13BD case, but they are similarly high in energy and unlikely to play any role at our collision energy. In addition to the detected radicals, theory suggests methyl thiophene + H_2 as a likely major product channel, as well as thioformaldehyde + 13BD, methyl thioketene + ethene, and thioketene + propene. These are exoergic by 16.5, 6.7, 46, and 4.4 kcal/mol, respectively. We plan a more detailed investigation of the isoprene reaction and its broader range of products in the near future.

To our knowledge, these $\text{S}(^3\text{P}) + \text{diene}$ reactions are the only atom–molecule reactions that have been shown to give cyclization to aromatic products under single collision conditions. Interestingly, sulfur dioxide also reacts with dienes, giving either cheletropic addition to form 5-membered sulfolane rings or hetero Diels–Alder reaction to form 6-membered sultine rings.⁴³ The former are thermodynamically favored and widely used industrially, while the latter can be formed at low temperature in solution and are of interest in

organic synthesis. Of greater relevance to our present work, the $\text{O}(^3\text{P})$ reaction with butadiene was recently studied with a similar combination of *ab initio*/TST theory and master equation modeling with crossed molecular beam scattering at 7.8 kcal/mol collision energy.³⁴ For the latter, the traditional rotatable mass spectrometer approach was used, providing universal detection for the myriad reaction channels observed. Although $\text{O}(^3\text{P})$ addition to the terminal or central carbon similarly initiates the reaction, a striking difference is the availability of several significantly exoergic product channels, with barriers far below the reactant energy, directly on the triplet surface: H_2CO with triplet diradical CH_2CHCH , H loss to form $\text{CH}_2\text{CHCHCHO}$, and vinyl + vinoxy radicals. There is a significant yield of these products found in both theory and experiment. ISC to the singlet surface is also clearly important, with CO + propene or HCO and allyl radical accounting for $\sim 60\%$ of the yield, again with theory and experiment in reasonable agreement. One point of disagreement is the observation of ketene + ethene in the experiment that was not seen in the modeling. This could perhaps be explained by the roaming reaction of vinyl + vinoxy on the triplet surface. No mention is made of the possible formation of furan or furanyl radicals, but it may be these are simply not competitive with the rapid loss via other pathways. On the other hand, $\text{H}_2 + \text{C}_4\text{H}_4\text{O}$ was detected in the experiment but assigned to vinyl ketene formation on the triplet surface. The difference between the $\text{S}(^3\text{P})$ and the $\text{O}(^3\text{P})$ PESs can be attributed almost entirely to the strong C–O bonds in all the products, 1.5–3 eV stronger than the corresponding C–S bonds. All the analogous product channels for sulfur are energetically closed except for the detected 2HT radicals, thiophene, and thioketene + ethylene.

One other point of interest raised here is the role of ISC in these reactions, a topic of some current interest. ISC has long been known to play a role in the reaction of $\text{O}(^3\text{P})$ with unsaturated hydrocarbons.^{18–21} As shown in the present study, it is usually found to take place via the initially formed triplet biradical minimum.²⁰ In a recent study of $\text{O}(^3\text{P}) + \text{cyclopentene}$, Ramasesha et al. showed that ISC is important in many distinct regions of the triplet PES.⁴¹ In recent work from our laboratory, we reported evidence of ISC in the exit channel of the reaction of $\text{O}(^3\text{P})$ with di- and trimethylamine.⁴⁴ In that case, initial H abstraction led to formation of a radical pair with a strong dipole–dipole interaction allowing time to permit ISC satisfying the El-Sayed rules for ultrafast ISC:⁴⁵ localized simultaneous change of orbital angular momentum and spin. In a recent study of $\text{O}(^3\text{P}) + \text{pyridine}$, evidence was seen for ISC in the entrance channel—the prereactive complex—giving a wide range of products not accessible via other regions of the triplet surface.⁴⁶ Calculations found evidence of an important frontier orbital similarly satisfying the El-Sayed rule. Although not discussed explicitly, the role of ISC is suggested by the products identified in the early $\text{S}(^3\text{P})$ studies of Gunning and co-workers discussed above and emphasized by McKee.³¹ In the present case, ISC at the triplet biradical minimum follows the “traditional” pathway seen for the $\text{O}(^3\text{P}) + \text{alkenes}$.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.3c01953>.

Experimental Methods, exploration and characterization of the PESs, properties of ISC regions in the $S(^3P) + 13BD$ reaction, MESS input to calculate RRKM rate coefficients, forward rate coefficients from 3CA RRKM calculations including the alternative path toward $T + H_2$, and branching fractions (PDF)

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Notes

The authors declare no competing financial interest.

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